Team Worksheet: Bestari Research Portal

SECJ2203 Software Engineering | Modules 3 and 4 | One copy per team | Dr. Mohamad Haider

How to use it. The Scrum Master makes one copy in Google Docs or OneDrive and gives the team edit rights. Type straight into the tables. Each developer fills in their own requirement card (R1 to R4) at the same time. Sheet P is for prototype pictures. At 4:20 the SM posts the link in the Webex chat.

Sheet 1: Team and sprint

Team: Sigma Aura **Members:** Nasaaie, Izzati, Thurga, Puteri, Farah, Najla

Role	Name	Requirement (R1 to R4)
Scrum Master	Nasaaie Bin Noriskamar	N/A
Developer 1	Nurul Izzati Liyana Binti Hashim	R1
Developer 2	Thurga Devi A/P Ponnambalam	R2
Developer 3	Puteri Anis Annisa Binti Mat Lazim	R3
Developer 4	Farah Adilah Binti Azman	R4
Developer 5	Najla Auni Binti Mohamad Asri	Sheet Q

Business problem statement. Who has what problem, why it matters to BRI, and how we will know it is solved. Write it together before anyone draws. (Template on page 4 of the Student Pack.)

Our business problem statement
Staff at the BRI Research Office faced challenges in managing research forms, which were handled manually via paper-based processes and email. This resulted in delays in the approval of research registrations, late submission of progress reports and the duplication of publication records. These issues were resolved through a portal that enables 100% of registrations and reporting to be conducted online, reducing approval processing time by at least 50% and eliminating duplicate records.

Sprint Goal (one sentence, written in Sprint Planning)

Our Sprint Goal
Develop a digital workflow prototypes R1 through R4 to eliminate overlapping publication records and ensure the timely submission of progress reports.

Stand-ups (2 minutes each)

Time	What is done?	What is blocking us?
3:15	Each member has completed the Personas and User Stories for their respective requirement cards (R1 to R4). Drafts for the Functional Requirements (FR) and Non-Functional Requirements (NFR) have been started.	Deciding how the registration number should be uniquely formatted across all four modules.
3:45	Question Tickets and the Assumption Log have been updated.	None.

Requirement card R1: Research Registration

Owner: Nurul Izzati Liyana Binti Hashim **Main user:** Researchers and Research Office staff

1. The business problem. Complete this before you draw anything.

Question	Our answer
Who has the problem?	Researchers and Research Office staff
What goes wrong today?	Researchers face the problem of not knowing the status of their registration, and need to wait for manual approval of their research application. The Research Office spends a lot of time chasing people to complete the approval process.
Why does it matter to BRI? (cost, delay or risk)	This matter caused delay, time, and energy consuming rather than a digitized method.
How will we know it is solved? (a measure)	A system that can easily help researchers to register their research projects in the portal, and once submitted, they can view their registered information in the portal, as well as the status.

	As for the Research Office, they shall be able to get the approval from each approval from the portal itself rather than fetching manual approval.

2. Persona and user stories. One typical user. The "so that" of each story must connect to the problem.

Name and role	Goal	Frustration today
Ana Smith, a Researcher	To submit a registration request to conduct a research about Human Rights and want to conduct it as soon as possible.	Has no visibility of his registration request whether it has been accepted, and left with frustration waiting for the Research Office to notify her about the result.

User stories: As a ..., I want to ..., so that ...
As a Researcher, I want to be able to view my registration status, so that I have more visibility about my registration request.
As a Research Office, I want to be able to complete the approval process swiftly and efficiently, so that the approval process can be completed faster with less hassle.

3. Requirements and how we will test them. At least 3 FR and 1 NFR. Source: a ticket number, or A for assumed.

ID	Requirement (shall ...)	FR / NFR	Test: how will we know it works?	Source
R1.1	The researcher shall be able to submit a new research project registration online.	FR	Confirm that by clicking the “Submit” button, a new successful saved request would appear in the system, with the status of “Pending Approval”.	A
R1.2	The system shall not allow the researcher to submit an incomplete registration form that is missing mandatory fields	NFR	Verify that by clicking “Submit” with empty mandatory fields, the system shall display an error	A

			message and prevent further action until the fields are completed.	
R1.3	The Head of Department, the Director of Research and the CEO shall be able to login to the portal with access to approve or reject each request.	FR	Confirm that all three of them have special access to the system, and can click on the “Approve” or “Reject” for each registration request issued.	A and R1
R1.4	The Research Office shall be able to see the status update of each approver on their screen.	FR	Verify that the Research Office screen has the features that show all three approval status.	A and R1
R1.5	The researcher shall be able to view the real-time status update of their registration.	FR	Verify that researchers will be able to view the status of all their submitted research registration.	A

Requirement card R2: IP Registration

Owner: Thurga Devi A/P Ponnambalam **Main user:** Research Office staff

1. The business problem. Complete this before you draw anything.

Question	Our answer
Who has the problem?	Research Office staff who keep IP records and researchers who own the IP.
What goes wrong today?	IP information is stored in separate files,making it difficult for staff to search,update and keep IP records.
Why does it matter to BRI? (cost, delay or risk)	It causes delays,increases the risk of missing or outdated IP information, and makes it harder to prepare accurate reports for management.
How will we know it is solved? (a measure)	Research Office staff can quickly search and update IP records in one central portal without needing to check multiple files.

2. Persona and user stories. One typical user. The "so that" of each story must connect to the problem.

Name and role	Goal	Frustration today
Hafiz bin Ismail, Research Office staff	To quickly search and maintain accurate IP records.	Check several spreadsheets by hand to find one record

User stories: As a ..., I want to ..., so that ...
1. As a staff member, I want to check IP records and make updates as required,so that the information is always up-to-date and accessible.
2. As a staff member, I want to get a warning before the expiry of an IP, so that we are able to renew it before losing protection.

3. Requirements and how we will test them. At least 3 FR and 1 NFR. Source: a ticket number, or A for assumed.

ID	Requirement (shall ...)	FR / NFR	Test: how will we know it works?	Source
R2.1	The Research Office personnel will have the capability to record an IP entry that will include IP number,end point date,quantity among other details	FR	Confirm that all the IP date has been entered and saved.The output will be a freshly created record in the system	A
R2.2	The Research Office staff would have the ability to search for an IP record based on the relevant IP details provided to them.	FR	Input the existing IP number into the search field. The resultant IP record will appear.	A
R2.3	The Research Office staff would be permitted to modify an IP record	FR	Choose an existing IP record, modify it and save it. The modified information will be displayed.	A
R2.4	The system should display results of the IP search within 2 seconds after the search is executed.	NFR	Look for an existing IP record in the system and check how much time it takes to respond.	A

R2.5	The Research Office staff shall be able to view the details of an existing IP record.	FR	Select an existing IP record from the search results. The system displays its IP number, research registration number, expiry date, quantity and related details.	A

Requirement card R3: Research Management

Owner: Puteri Anis Annisa binti Mat Lazim **Main user:** Researcher and Research Office staff

1. The business problem. Complete this before you draw anything.

Question	Our answer
Who has the problem?	Researcher and Research Office staff
What goes wrong today?	Progress reports (Form RO-04) are submitted on paper and often late as it requires manual phone call reminders and manual data entry into spreadsheets.Beside that, the research statuses,team details and progress reports sit in separate files which makes the researcher and staff cant see it in one same layout.
Why does it matter to BRI? (cost, delay or risk)	Research office staff spend excessive time manually chasing overdue reports.Management lacks real-time visibility into project delays or risks which takes days to compile for management reporting.
How will we know it is solved? (a measure)	All the progress reports are tracked digitally with automated notifications,reducing report submission delays and allowing staff to view any project status within a minute.

2. Persona and user stories. One typical user. The "so that" of each story must connect to the problem.

Name and role	Goal	Frustration today
Dr. Aisha Rahman (Researcher)	Easily track project progress,update team status and submit progress reports on	Has to fill out a paper progress report and receives phone

	time without manual follow-up calls.	reminders when the deadline passes unnoticed.

User stories: As a ..., I want to ..., so that ...
As a researcher, I want to view and update my research project current status and progress online, so that I don't need to fill in the form in paper and submit it to the office manually.
As a Research Office staff member, I want to see an automated list of overdue annual progress reports and send automated reminders, so that other staff do not spend time chasing researchers over the phone to complete the form and I do not need to go through a lot of files to find the recent project updates.

3. Requirements and how we will test them. At least 3 FR and 1 NFR. Source: a ticket number, or A for assumed.

XZZX	Requirement (shall ...)	FR / NFR	Test: how will we know it works?	":;
R3.1	The researcher shall be able to update their research project status,team members and progress report details online.	FR	Login as researcher, update recruitment numbers on projects and verify system saves changes instantly.	RO-04
R3.2	The notification service shall be able to send automated reminders to all researchers for their upcoming and overdue annual progress reports.	FR	Set system update and check that an automated reminder email has been sent to the researcher.	A
R3.3	There should be a generated centralized dashboard view listing like active , on-hold or completed and terminated research projects for research office staff.	FR	Login as Research Office Staff and verify all projects listed with their current status.	Brief R3
R3.4	The staff shall be able to retrieve a summary list of all overdue annual progress reports within 2	NFR	Filter for overdue progress reports on the dashboard and record that results display in under 2 seconds.	A

	seconds from submitting the filtered query.			
R3.5	The system should be able to restrict the project's status modification access to only the principal investigator and the research office staff only while other staff can only be in view mode only and cannot edit them.	FR	Login as standard staff and try to edit the project's status.	A

Requirement card R4: Publication

Owner: Farah Adilah Binti Azman **Main user:** BRI Research Office staff and management

1. The business problem. Complete this before you draw anything.

Question	Our answer
Who has the problem?	BRI Research Office staff and management.
What goes wrong today?	Publications records were stored in different files, making it difficult to access and have duplicates.
Why does it matter to BRI? (cost, delay or risk)	Staff spend more time searching for records leading to the increasing risk of duplicate records and inaccurate record sheets which can cause loss to the company.
How will we know it is solved? (a measure)	All publication records will be stored in one centralized portal making them easier to access and manage.

2. Persona and user stories. One typical user. The "so that" of each story must connect to the problem.

Name and role	Goal	Frustration today
BRI Research Office Manager	To access and manage publication records efficiently.	Publication records are scattered across different files making it difficult to verify the information.

User stories: As a ..., I want to ..., so that ...

As a Management user, I want to access all publication records through one centralized portal, so that I can easily retrieve and review publication information without searching through multiple files.
As a Management user, I want the system to compile the publication records and help to identify duplicates, so that I can maintain an accurate and consistent publication record sheet.

3. Requirements and how we will test them. At least 3 FR and 1 NFR. Source: a ticket number, or A for assumed.

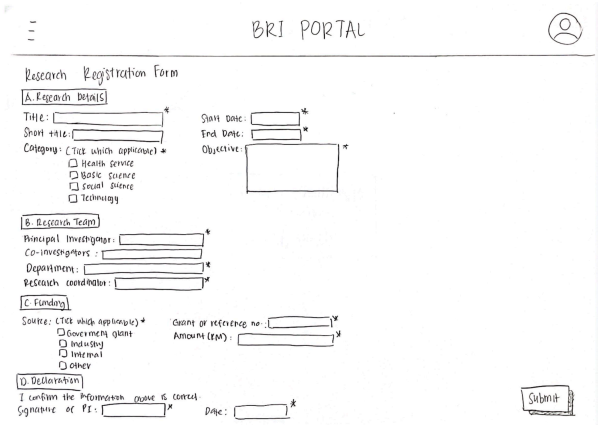
ID	Requirement (shall ...)	FR / NFR	Test: how will we know it works?	Source
R4.1	The system shall store all publication records in a centralized database.	FR	Verify that the publication records submitted through the portal are stored in the central database.	A
R4.2	The system shall allow management to view all publication records in one portal.	FR	Log in as management and verify that records from different sources can be accessed in one place.	A
R4.3	The system shall allow management to search and retrieve records by publication title or author.	FR	Search using a publication title or author and verify that the relevant records are displayed.	A
R4.4	The system shall detect duplicate publication records before saving new records.	FR	Detects duplicate when records submitted have matching publication details.	A
R4.5	The system shall display search results within 3 seconds under normal operating conditions.	NFR	Search multiple records and the result for each record will be displayed within 3 seconds.	A

Sheet P: Prototypes

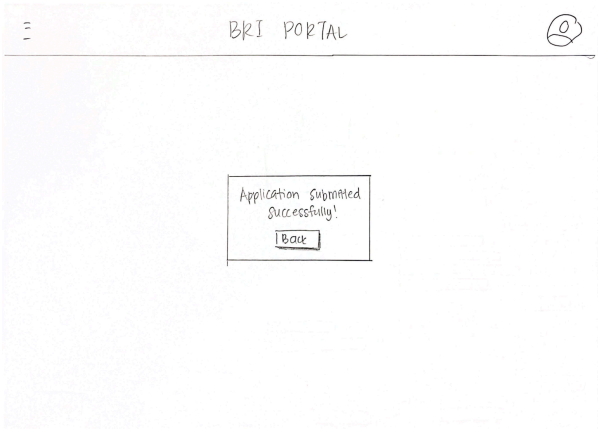
Paste a photo of your paper sketch or a screenshot of your drawing. Label every field, button and message, and give an example value for each field. Use the same name for information that passes between requirements, for example the registration number.

R1 Research Registration	R2 IP Registration

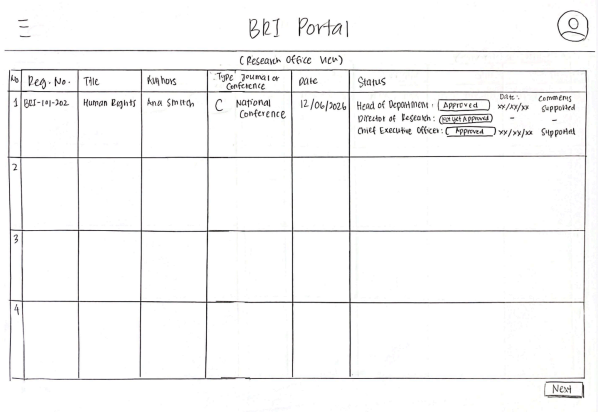
Registration Form

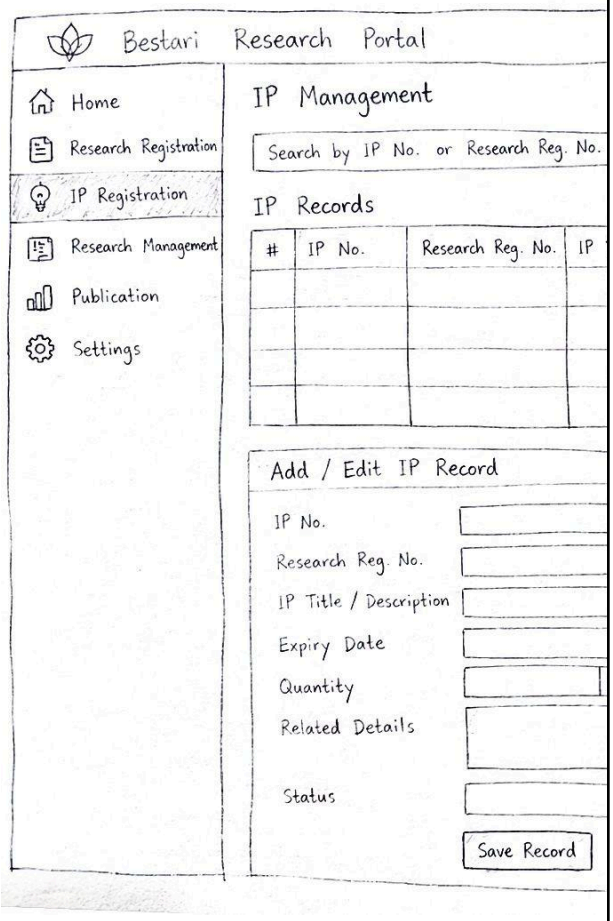


Successful Registration Request Message

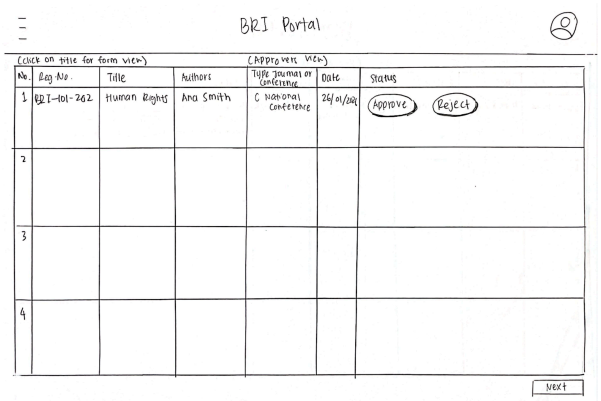


Application List View for Researcher Office

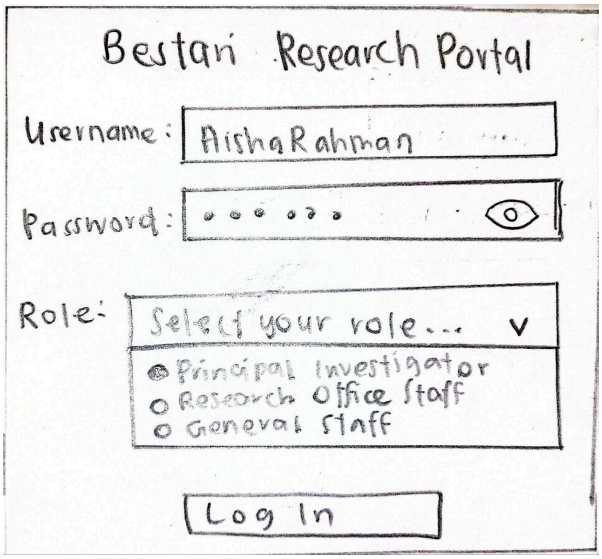


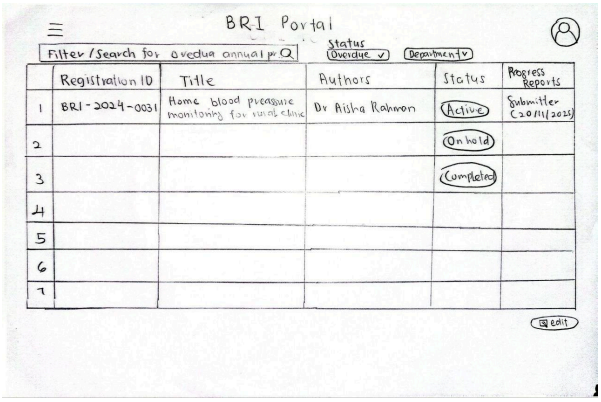


Application List View for Approvers

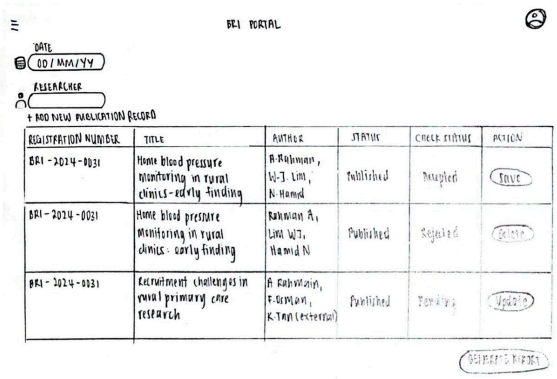


R3 Research Management





R4 Publication



Sheet Q: Question tickets and assumptions

Team: Sigma Aura **Members:** Nasaaie, Izzati, Thurga, Puteri, Farah, Najla

Tickets. You have **5 tickets** for the whole session: at most 3 per visit, at most 2 visits. Write each question **before** you visit. Write the answer exactly as the client says it.

Ticket	Visit (1 or 2)	For R #	Question	Answer (the client's words)
1	1	R1	Form RO-01 currently requires three separate signatures (Head Of Department, Director, Chief Executive Officer). Will all three approvers need full review and approval access inside the portal?	The Head Of Department will approve the registration and after approval the director will handle it. Every approval should be recorded in the system.
2	2	R4	On Form RO-07, the same paper was listed twice (rows 1 and 2). How should the portal prevent duplicate publication entries when researchers submit their records?	The system must check the title or publication number. If the same title is already in the system, show a warning and cannot save a duplicate entry.
3	2	R3	On Form RO-04, progress reports were received 15 days late. Should the portal send automated email reminders before the due date, and block late submissions?	Send an email reminder 3 days before the due date. If late, researchers still can submit but the system must show red status ‘Late’ to admin.
4	2	R2	For IP records (R2), should every IP record be strictly linked to an approved research registration number, or can independent IP be added?	Must link to approved research number first. Cannot add independent IP without research registration number.
5	1	R3	When a project status changes to 'On hold' or 'Terminated' on Form RO-04, should the portal automatically suspend progress report schedules and notify management?	The researcher or coordinator will change the status. If the status is changed, we do not have to inform the Head Of Department.

Assumption log. Write down everything you decided **without asking**, including what you think a term in the brief means. A logged assumption that is later confirmed or corrected earns points. One you never noticed earns nothing.

#	What we assumed	Why we assumed it	R #	Confirmed? (ticket # or No)
1	Approvals must follow a strict 3-tier sequence (Head Of Department → Director → Chief Executive Office) as shown on RO-01 signature block.	Because layout of approval section E in Form RO-01.	R1	Ticket 1
2	Duplicate publication entries occur because yearly email submissions lack automated Digital Object Identifier (DOI) deduplication.	Because evidence of a duplicate conference paper entry on Form RO-07.	R4	Ticket 2
3	Progress report deadlines (RO-04) are calculated based on the project's start date rather than a single fixed calendar date.	Because the dates on the form match the project start date.	R3	Ticket 3
4	Changing project status to 'On hold' or 'Terminated' requires notifying the Head Of Department immediately.	Because we thought big changes always need boss approval	R3	Corrected (Ticket 5: No need to inform Head Of Department)
5	IP assets come straight from ongoing research grants.	Because research groups always make new items from funded work.	R2	Ticket 4
6	All module records must use a standardized registration number format.	Because standardized registration format used across Forms RO-01, RO-04, and RO-07.	ALL	Confirmed (RO-01/04/07)

Sheet C: Fit check, change and retrospective

Team: Sigma Aura **Members:** Nasaaie, Izzati, Thurga, Puteri, Farah, Najla

Part 1. Do the four prototypes fit together? (before 3:55)

Pick pairs of requirements. What information must pass from one to the other? Is it shown in both prototypes, under the same name?

From R# to R#	What information must pass between them?	In both prototypes?	What we fixed
R1 to R3	Research registration number and project title	Yes	Standardized the field label to research registration number on both screens.
R1 to R4	Research registration number to link publications directly to research projects and prevent orphan records.	Yes	Added mandatory research registration number field in publication entry screen.
R1 to R2	Research registration number to associate IP assets with approved research grants	Yes	Ensured IP search bar and details view display research registration number.

Part 2. Change request (3:55)

Change received: The approval process remains the same and cannot be completely digitalised, as it requires the approval from all three either by signature or replying to an email. They shall not be granted access to the portal, and the approval updates for each of them are recorded by the Research Officer. The researcher shall be able to see the status of who have approved so far rather than the final complete status like previously.

Question	Our answer
Which requirement and prototype change, and how?	The requirements change at the Registration requirement, whereas all the approval must not be done digitally in the portal, instead, must be continued to be done manually in paper form.
Which other requirements are affected (knock-on effects)?	Research Management is affected because annual progress report submission and active project tracking must remain locked or labeled as “Pending Approval” until the Researcher Officer logs that all three approvals are completed in the portal.
What would we ask the client before building this change?	Since approvals can happen in any order or at the same time to save time, should the Research Officer be able to record

	approvals in any sequence or does the portal still require a specific order?

Part 3. Retrospective (while others present)

Question	Our answer
What did we assume without realising it, and how did we find out?	We assumed higher management (Head of Department, Director, Chief Executive Officer) would log into the portal to review and approve research registrations online. We found out this was wrong when Change Request 01 was issued, stating they travel often and will only approve via paper signature or email outside the portal.
If we had only half the time, which two requirements would we build first, and why? (M4 slide 36)	Research Registration and Research Management, because research registration is the mandatory starting point for all research work and tracking progress reports directly fixes Bestari Research Institute's problem with late 15-day submissions.
One thing we would change about how we asked questions	We would ask early questions about user roles and constraints before designing user interfaces for them such as whether executive management would actually use portal accounts.